

Real-time Superquadric Representation of Cuboids and Cylinders using Instant Surface Normal Map for Dynamic Object Integration in AR

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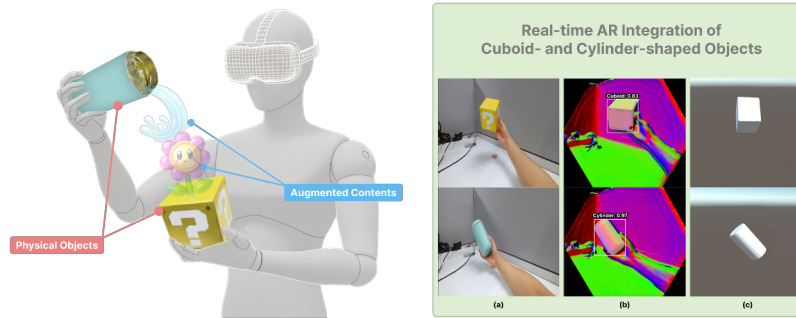


Figure 1: Overview of the proposed pipeline for real-time AR integration of cuboid- and cylinder-shaped objects. (a) RGB-D input frames capturing real-world objects. (b) Instant Surface Normal (ISN) maps are used for object detection and segmentation. (c) Superquadric-based 3D representation of cuboids and cylinders, enabling their seamless incorporation as augmented content in AR environments.

ABSTRACT

We present a real-time AR system that enables interaction with physical cuboid- and cylinder-shaped objects by representing them as compact superquadrics. Our approach uses Instant Surface Normal (ISN) maps—color images encoding per-pixel geometry—to detect and segment objects based on shape rather than appearance. Superquadric parameters are extracted sequentially, allowing efficient and updatable 3D modeling in real time. Experimental results show accurate detection, consistent segmentation, and reconstruction quality comparable to existing methods, while achieving over 5× faster inference. This system allows physical objects to be seamlessly integrated as interactive elements in dynamic AR environments.

Index Terms: Augmented Reality, Superquadric Representation, Object Detection, Surface Normal Map.

1 INTRODUCTION

Despite rapid advances in hardware, current augmented reality (AR) systems remain limited in their ability to recognize and model physical objects in the environment. Most systems reliably detect only basic elements, such as planar surfaces or hands, treating surrounding objects as undifferentiated obstacles rather than individual entities, restricting the natural interaction between virtual and physical worlds.

Existing approaches, including SLAM-based reconstruction [3, 11], radiance field representations [5, 9], and user-driven modeling [6, 10], struggle to meet key AR requirements such as compactness, updateability, and real-time performance. These limitations make it

difficult to integrate physical objects into interactive AR applications.

To address this challenge, we present a unified real-time system that represents cuboid- and cylinder-shaped physical objects as compact superquadrics [1]. Our approach leverages Instant Surface Normal (ISN) maps—geometry-driven color images derived from depth data—to enable appearance-independent detection, efficient segmentation, and real-time superquadric parameter extraction. By focusing on lightweight primitive representations, the proposed system enables AR applications to robustly virtualize and interact with physical objects in real time.

2 REAL-TIME PRIMITIVE REPRESENTATION

Fig. 1 provides an overview of the proposed system for real-time primitive-based object representation in AR. The system takes a depth input from an RGB-D sensor and converts it into an Instant Surface Normal (ISN) map, a color image that encodes geometric information per-pixel. Unlike conventional RGB images, the ISN map emphasizes surface orientation and geometry, enabling object perception based on its geometric shapes rather than texture or appearance.

By training the off-the-shelf detection model (specifically YOLOv8 [4]) with the synthetic ISN map data set of cuboid and cylinders, the system can detect arbitrary objects that have cuboid and cylinder-shapes. The detected objects are then efficiently segmented using ISN-specific geometric characteristics, enabling stable instance segmentation while maintaining real-time throughput.

The segmented objects are then converted into superquadric representations [1] by sequentially estimating geometric parameters, including rotation, scale, and translation. This estimation process is non-iterative and lightweight, avoiding costly optimization procedures while preserving sufficient geometric accuracy.

By combining geometry-driven perception with compact superquadric representations, the system maintains real-time performance and is able to support continuous updates of the object pose and position. This design allows physical objects to be treated as interactive entities in AR applications, enabling consistent integration of real-world objects into dynamic virtual environments.

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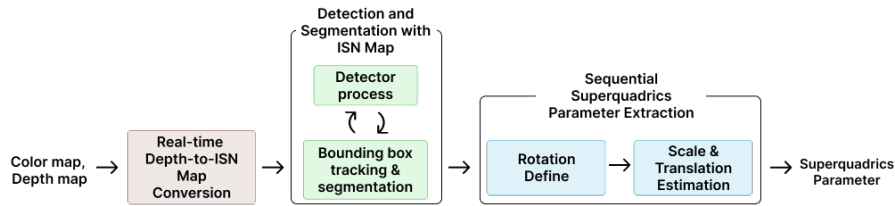


Figure 2: System architecture of the proposed method. ISN maps derived from depth frames enable real-time detection, segmentation, and sequential superquadric parameter extraction in a non-iterative pipeline.

3 RESULTS AND DISCUSSION

We evaluated the proposed system through three experiments, each designed to assess its detection robustness, segmentation efficiency, and the quality of superquadric-based object representation. As data sets for the following three experiments, we used eight total cases consisting of four objects (cracker box, wood block, chips can, and pitcher) and two camera view points (N1 and N2) from the YCB data set [2].

From the evaluation of the robustness of cuboid and cylinder detection trained with synthetic ISN images of cuboids and cylinders, the detector achieved consistent performance in all test cases, with mAP@0.5 values above 0.6 and several cases exceeding 0.95. These results demonstrate that ISN maps enable robust geometry-driven object detection that generalizes across viewpoint and scale variations, while remaining independent of appearance cues such as the color of the object. Performance variations from different cases shown in the experiment suggest that increasing the pose and scale diversity in synthetic ISN images used to train the model could further improve the robustness of the detection.

We compared the proposed ISN-based asynchronous instance segmentation with a vanilla YOLOv8 baseline to evaluate segmentation stability and speed. The ISN-based method achieved comparable or higher segmentation accuracy in all cases, with mean IoU values typically between 0.49 and 0.57, while consistently maintaining real-time performance below 0.021s per frame. In cases where the baseline showed marginally higher accuracy, the difference was negligible (less than 0.005 mIoU). Overall, these results indicate that ISN-based asynchronous segmentation provides a more stable and robust solution for real-time AR object representation.

We evaluated the proposed sequential extraction of superquadric parameters against representative baselines based on optimization [8] and deep learning methodology [7]. The proposed method achieved reconstruction accuracy comparable to state-of-the-art deep learning approaches, with a Chamfer distance of approximately 0.025 and a volumetric IoU of 0.17, while significantly reducing computational cost. In particular, our method required only 11.5 ms for superquadric parameter extraction, achieving more than 5x faster inference speed compared to the existing method. Combined with real-time segmentation, the entire system operates at approximately 33 ms per frame, enabling continuous real-time object representation suitable for interactive AR applications.

4 FUTURE WORK AND CONCLUSION

We presented a real-time system for representing cuboid- and cylinder-shaped physical objects as compact superquadrics for interactive AR applications. Using Instant Surface Normal maps, the proposed approach enables geometry-driven detection, efficient segmentation, and lightweight object modeling without relying on computationally expensive optimization. Although the system is limited to primitive-shaped objects, this design choice allows stable real-time performance and compact representations suitable for object-aware AR. These trade-offs motivate future work toward

learning-based extensions that preserve the advantages of primitive representations while supporting more complex object geometries and end-to-end optimization, further advancing the integration of physical objects as interactive entities in dynamic AR environments.

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